

Create the K8s EKS, further you have to do the deployment of the Nginx application and access the application outside the cluster.

Created EKS cluster with t3.micro 5 nodes

```
root@SriJay:/home/vicky# kubectl get nodes
NAME                                STATUS    ROLES    AGE    VERSION
ip-192-168-22-189.ap-south-1.compute.internal Ready    <none>   13m    v1.34.2-eks-ecaa3a6
ip-192-168-25-53.ap-south-1.compute.internal Ready    <none>   13m    v1.34.2-eks-ecaa3a6
ip-192-168-47-18.ap-south-1.compute.internal Ready    <none>   13m    v1.34.2-eks-ecaa3a6
ip-192-168-69-1.ap-south-1.compute.internal Ready    <none>   13m    v1.34.2-eks-ecaa3a6
ip-192-168-77-158.ap-south-1.compute.internal Ready    <none>   13m    v1.34.2-eks-ecaa3a6
root@SriJay:/home/vicky# kubectl get deployment
NAME          READY   UP-TO-DATE   AVAILABLE   AGE
nginx-deployment 1/1     1            1           3m10s
root@SriJay:/home/vicky# kubectl get pods
NAME                                STATUS    RESTARTS   AGE
nginx-deployment-7457467ffd-g5x8b 1/1      Running    0           3m18s
root@SriJay:/home/vicky# kubectl get svc nginx-service
NAME    TYPE        CLUSTER-IP    EXTERNAL-IP    PORT(S)
nginx-service LoadBalancer 10.100.172.42  a0766f79f31594fac9ab0dd8418db8bd-52789323.ap-south-1.elb.amazonaws.com 80:31310/TCP
root@SriJay:/home/vicky#
```

Created with deployment image for nginx

```
root@SriJay:/home/vicky# cat nginx-deployment.yaml
apiVersion: apps/v1
kind: Deployment
metadata:
  name: nginx-deployment
spec:
  replicas: 2
  selector:
    matchLabels:
      app: nginx
  template:
    metadata:
      labels:
        app: nginx
    spec:
      containers:
      - name: nginx
        image: nginx
        ports:
        - containerPort: 80
```

Created nginx service yaml file

```
ContainerPort: 80
root@SriJay:/home/vicky# cat nginx-service.yaml
apiVersion: v1
kind: Service
metadata:
  name: nginx-service
spec:
  type: LoadBalancer
  selector:
    app: nginx
  ports:
  - port: 80
    targetPort: 80
```

Created load balancer service and exposed on port 80

```
root@SriJay:/home/vicky# kubectl get svc nginx-service
NAME          TYPE          CLUSTER-IP      EXTERNAL-IP      PORT(S)
nginx-service  LoadBalancer  10.100.172.42    a0766f79f31594fac9ab0dd8418db8bd-52789323.ap-south-1.elb.amazonaws.com  80:31310/TCP
2m44s
```

Accessing the application from browser

